

The mass gap as a finite list of theorems: the Hamiltonian lattice route

On the Hamiltonian lattice route the Jaffe–Witten conjecture decomposes into four named statements, and their present status is: (T1) every finite periodic lattice has a positive gap $\Delta_{a,L}$ above a unique gauge-invariant ground state, proved below with all constants explicit; (T2) at strong coupling the gap stays bounded below uniformly in the volume, established in the Euclidean form by the Osterwalder–Seiler cluster expansion and cited here with its hypotheses, while the naive Kato bound fails because the magnetic term is extensive; (T2′) the gap of the infinite lattice stays positive for every coupling, which is the statement that four-dimensional non-abelian lattice gauge theory has no Coulomb phase, open, and false for the abelian group by Guth and Fröhlich–Spencer; (T3) along the asymptotic-freedom curve $g(a)$ the infinite-volume gap Δ_a satisfies $\Delta_a/(\hbar c \Lambda(g(a))) \rightarrow m/\Lambda \in (0, \infty)$, open, and this single limit is the Millennium content once the theory exists; (T4) the continuum, infinite-volume theory with the Osterwalder–Schrader axioms, known in finite volume with an ultraviolet cutoff removed (Balaban; Magnen–Rivasseau–Sénéor) and open beyond. The small-volume expansion controls one corner: for $L \ll \hbar c/\Lambda$ the gap is $\delta_1 g(L)^{2/3} \hbar c/L$ (C133, Lüscher), so the dimensionless gap $z(L) = \Delta(L)L/(\hbar c)$ must cross over from $\propto g^{2/3}$ at small L to linear growth $mL/(\hbar c)$ at large L . The first theorem-sized target this leaves is T2 in Hamiltonian form: a volume-uniform gap for $g \geq g_0(N)$, whose radius g_0 is the number the entire problem asks to push to zero. Sources are cited inline with reading labels; nothing here is promoted.

1. The finite-lattice Hamiltonian

Let G be a compact connected simple Lie group with Lie algebra basis T^a normalized by $\text{tr } T^a T^b = \frac{1}{2} \delta^{ab}$ in a fixed faithful representation (for $SU(N)$ the fundamental). Let $\Lambda = (a\mathbb{Z}/Na\mathbb{Z})^3$ be the periodic spatial lattice with spacing a , side $L = Na$, link set $\mathcal{E}$ and plaquette set $\mathcal{P}$; $|\mathcal{E}| = 3N^3$, $|\mathcal{P}| = 3N^3$. The Hilbert space is $\mathcal{H} = L^2(G^{\mathcal{E}}, \text{Haar}^{\otimes \mathcal{E}})$. On the copy of G attached to a link ℓ , the electric field components E_ℓ^a are $-i$ times the left-invariant vector fields, and $\sum_a (E_\ell^a)^2 = -\Delta_\ell$ is the Laplace–Beltrami operator of the bi-invariant metric, whose eigenvalues on the Peter–Weyl blocks are the quadratic Casimirs $C_2(R)$ with multiplicity $(\dim R)^2$. The Kogut–Susskind Hamiltonian (Phys. Rev. D 11 (1975) 395, metadata) in the convention used here is

$$H = \frac{\hbar c}{a} \left[\frac{g^2}{2} \sum_{\ell \in \mathcal{E}} (-\Delta_\ell) + \frac{2}{g^2} \sum_{p \in \mathcal{P}} (N - \text{Re tr } U_p) \right] = \frac{\hbar c}{a} [g^2 K + g^{-2} V],$$

with g dimensionless, U_p the ordered product of the four link variables around p , and K, V the dimensionless electric and magnetic operators. Other conventions rescale g^2 by a fixed number. The dimensionless spectrum of $aH/(\hbar c)$ depends on g, N and G only:

$$\Delta_{a,L} = \frac{\hbar c}{a} \delta(g; N, G).$$

Gauge transformations $\gamma : \Lambda \rightarrow G$ act by $(\mathcal{V}_\gamma \psi)(U) = \psi(\gamma_x U_\ell \gamma_y^{-1})$ for $\ell = (x, y)$; they are unitary, commute with H (the Laplacian is bi-invariant and $\text{tr } U_p$ is conjugation invariant) and preserve positivity of functions. The physical space $\mathcal{H}_{\text{phys}}$ is the joint fixed space (Gauss’s law).

2. T1: every finite lattice has a gap above a unique physical ground state

Theorem 1. For every $a > 0$, $N \geq 2$, $g > 0$ and G as above:

1. H is self-adjoint, bounded below, and has compact resolvent; its spectrum is a sequence $E_0 < E_1 < \dots \rightarrow \infty$ of eigenvalues of finite multiplicity.

2. E_0 is simple with a strictly positive eigenfunction ψ_0 , so $\Delta_{a,L} = E_1 - E_0 > 0$.
3. $\psi_0 \in \mathcal{H}_{\text{phys}}$, $\mathcal{H}_{\text{phys}}$ reduces H , and the physical gap satisfies $\Delta_{a,L}^{\text{phys}} \geq \Delta_{a,L} > 0$.

Proof. (1) $-\Delta_\ell$ is essentially self-adjoint on $C^\infty(G)$, nonnegative, with compact resolvent, since G is a compact manifold and Peter–Weyl gives the eigenvalues $C_2(R)$ with finite multiplicities and $C_2(R) \rightarrow \infty$ along any enumeration of irreducible representations. $K = \sum_\ell (-\Delta_\ell)$ acts on the finite tensor product with eigenvalues $\sum_\ell C_2(R_\ell)$; for each bound E only finitely many assignments $\ell \mapsto R_\ell$ have $\sum_\ell C_2(R_\ell) \leq E$, so K has compact resolvent. V is multiplication by a continuous function with $0 \leq V \leq 2N|\mathcal{P}|$, hence bounded, and a bounded perturbation of an operator with compact resolvent has compact resolvent by the resolvent identity $(H - z)^{-1} = (H_0 - z)^{-1}[1 + (g^{-2}V)(H_0 - z)^{-1}]^{-1}$ for z far in the negative axis. Discreteness with finite multiplicities and $E_n \rightarrow \infty$ follow, and E_0 is an eigenvalue.

- (2) The heat kernel of $-\Delta$ on a compact connected Riemannian manifold is strictly positive for $t > 0$ (standard, from the strong maximum principle or the Peter–Weyl expansion; metadata level), so the kernel of e^{-tK} on $G^\mathcal{E}$, a product of strictly positive kernels, is strictly positive. With Brownian motion X_t on $G^\mathcal{E}$ generated by K (time rescaled by $g^2\hbar c/a$), the Feynman–Kac formula gives

$$(e^{-tH}\psi)(U) = \mathbb{E}^U \left[\exp \left(-\frac{\hbar c}{ag^2} \int_0^t V(X_s) ds \right) \psi(X_t) \right],$$

valid because V is bounded and continuous. For $\psi \geq 0$, $\psi \neq 0$, the weight is at least $e^{-thc \cdot 2N|\mathcal{P}|/(ag^2)} > 0$ and the law of X_t has a strictly positive density, so $(e^{-tH}\psi)(U) > 0$ for every U : e^{-tH} is positivity improving. Reed–Simon IV, Theorem XIII.44 (metadata; number corroborated as in B78) then gives that E_0 is simple with $\psi_0 > 0$ almost everywhere, and $E_1 > E_0$ because E_1 is the next eigenvalue of a discrete spectrum.

- (3) $\mathcal{V}_\gamma\psi_0$ is a normalized eigenfunction for E_0 , hence $\mathcal{V}_\gamma\psi_0 = c(\gamma)\psi_0$ with $|c| = 1$; both functions are positive, so $c(\gamma) = 1$ and $\psi_0 \in \mathcal{H}_{\text{phys}}$. The projection $P = \int \prod_x d\gamma_x \mathcal{V}_\gamma$ onto $\mathcal{H}_{\text{phys}}$ commutes with $(H - z)^{-1}$, so $\mathcal{H}_{\text{phys}}$ reduces H and the restricted operator has compact resolvent, ground state ψ_0 , and next eigenvalue at least E_1 . $\square$

Theorem 1 defines the object whose limits the conjecture is about. It proves nothing about those limits: $\Delta_{a,L}$ is $\hbar c/a$ times a number that may tend to zero as $N \rightarrow \infty$ or as $g \rightarrow 0$.

3. T2: strong coupling, and why volume uniformity is the whole difficulty

At $g \rightarrow \infty$ the operator g^2K dominates. Its ground state is the constant function, with eigenvalue 0, and its excitations are link representations. Gauge invariance excludes a single excited link: the lowest physical excitation of K is a closed flux loop, at minimum the four links of one plaquette in the lowest nontrivial representation, with electric energy $4 \cdot \frac{1}{2}C_2(R_{\min})g^2\hbar c/a = 2C_2(R_{\min})g^2\hbar c/a$; for $SU(N)$, $C_2(\text{fund}) = (N^2 - 1)/(2N)$. So the unperturbed physical gap is

$$\gamma_{\text{phys}}(g) = 2C_2(R_{\min})g^2 \frac{\hbar c}{a},$$

uniform in N because a single plaquette loop costs the same on every lattice.

The naive bound fails. The magnetic term satisfies $0 \leq g^{-2}V \leq 2N|\mathcal{P}|g^{-2}$ in operator norm, so Kato’s perturbation bound gives $\Delta^{\text{phys}} \geq \gamma_{\text{phys}} - 2\|g^{-2}V\|\hbar c/a = (2C_2g^2 - 4N|\mathcal{P}|g^{-2})\hbar c/a$, positive only when $g^4 > 2N|\mathcal{P}|/C_2$. The bound is extensive in the volume and useless as $N \rightarrow \infty$. Volume uniformity therefore requires using that V is a sum of $|\mathcal{P}|$ terms each of norm at most $2Ng^{-2}$ and each acting on four links: locality, not smallness of the norm.

What is established. Osterwalder and Seiler (Ann. Phys. 110 (1978) 440; abstract, B78) prove for the Euclidean lattice theory with any compact gauge group, at sufficiently strong coupling, reflection positivity, the existence of the infinite-volume limit and Wilson’s confinement bound by a cluster expansion; exponential clustering of gauge-invariant correlations uniform in the volume, that is a mass gap of the transfer matrix in units of $1/a$, is the standard consequence of the same expansion (not read at section level). Hamiltonian versions rest on gap-stability theorems for local perturbations of commuting, frustration-free Hamiltonians (Yarotsky, Commun. Math. Phys. 261 (2005) 799; Bravyi–Hastings–Michalakis, J. Math. Phys. 51 (2010) 093512; both metadata): $g^2 K$ is a sum of commuting link terms with a product ground state and a uniform local gap, and $g^{-2} V$ is a sum of bounded local terms. Their hypotheses assume finite-dimensional local spaces or bounded local terms, and $-\Delta_\ell$ is unbounded on the infinite-dimensional $L^2(G)$, so a Hamiltonian proof of T2 must either truncate each link to the representations with $C_2(R) \leq C_{\max}$ and control the truncation, or use the relative boundedness of V with respect to K . This is the first bounded theorem in the list:

T2 (Hamiltonian form, target). There exist $g_0(N) < \infty$ and $c(N) > 0$ such that for all $g \geq g_0$ and all lattice sizes $N \geq 2$, $\Delta_{a,L}^{\text{phys}} \geq c(N) g^2 \hbar c/a$.

4. T2’: no Coulomb phase at fixed cutoff

Fix a and g and let $N \rightarrow \infty$. Define $\delta_\infty(g) = \liminf_{N \rightarrow \infty} \delta(g; N, G)$. T2 says $\delta_\infty(g) \geq c g^2 > 0$ for $g \geq g_0$. The lattice form of the conjecture at fixed cutoff is

T2’. $\delta_\infty(g) > 0$ for every $g > 0$.

For $G = U(1)$ this is false: Guth (Phys. Rev. D 21 (1980) 2291) and Fröhlich–Spencer (Commun. Math. Phys. 83 (1982) 411) prove a massless Coulomb phase at weak coupling (abstract, B78), so $\delta_\infty(g) = 0$ below a critical coupling. T2’ for a non-abelian G is the statement that the transition the abelian theory undergoes does not occur, that is confinement at all couplings, which lattice computations support and no proof reaches. The heuristic of G07 §3 locates the difference: the abelian theory is all valley, while the non-abelian potential has flat directions of measure zero along which a transverse zero-point energy grows. T2’ is where that mechanism would have to be turned into an estimate at weak coupling.

5. T3: the scaling limit, as one asymptotic statement

Along the continuum curve the bare coupling runs with the cutoff. With $b_0 = 11N/(48\pi^2)$ and $b_1 = 34N^2/(3(16\pi^2)^2)$ for $SU(N)$ (the two-loop coefficients in the convention $\beta(g) = -b_0 g^3 - b_1 g^5 + \dots$; Gross–Wilczek, Politzer, abstract, B78), the lattice Λ parameter is defined by

$$a\Lambda_{\text{lat}}(g) = (b_0 g^2)^{-b_1/(2b_0^2)} e^{-1/(2b_0 g^2)} [1 + O(g^2)],$$

a dimensionless function that vanishes faster than any power of g .

T3. The limit $\frac{m}{\hbar c \Lambda_{\text{lat}}} = \lim_{g \rightarrow 0} \frac{\delta_\infty(g)}{a\Lambda_{\text{lat}}(g)}$ exists and lies in $(0, \infty)$.

This is the Jaffe–Witten mass gap m expressed on the lattice route: the dimensionless infinite-volume gap must vanish as $g \rightarrow 0$ exactly like $e^{-1/(2b_0 g^2)}$ times the prescribed power, with a positive finite prefactor. Any weaker vanishing gives $m = \infty$ in the continuum (the theory is gapped only at the cutoff scale, as the compact abelian theory in three dimensions is, G07 §4); any faster vanishing, or $\delta_\infty = 0$ at some g , gives $m = 0$. T3 presupposes T2’. The order of limits is $N \rightarrow \infty$ at fixed g first, then $g \rightarrow 0$ with $a = \Lambda_{\text{lat}}^{-1} a\Lambda_{\text{lat}}(g)$; Jaffe and Witten note (p. 6,

passage) that a gap uniform in the volume would also serve to construct the infinite-volume limit, which is the reverse order.

The small-volume corner. For $L \ll \hbar c/\Lambda$ the constant-mode sector of the continuum torus theory has gap $\delta_1 g(L)^{2/3} \hbar c/L$ (C133, G07 Proposition 7), and Lüscher’s expansion (Nucl. Phys. B219 (1983) 233, abstract) supplies the corrections in powers of $g^{2/3}$ with g the renormalized coupling at scale L . In the variable $z(L) = \Delta(L)L/(\hbar c)$ the two ends read

$$z(L) = \delta_1 g(L)^{2/3} [1 + O(g^{2/3})] \quad (L \rightarrow 0), \quad z(L) \sim \frac{mL}{\hbar c} \quad (L \rightarrow \infty),$$

and Lüscher–Münster (Nucl. Phys. B232 (1984) 445, abstract) place the crossover near $z \simeq 2$. The conjecture, in this variable, is that $z(L)$ is a function that starts at zero and grows without bound, with m the slope of its linear asymptote. The corner is the only place where the gap of the continuum theory is presently computable, and it is controlled by the mechanism of G08.

6. T4: existence

The lattice route needs the continuum limit of the Euclidean lattice theory with Osterwalder–Schrader reconstruction. Known: ultraviolet stability of four-dimensional lattice Yang–Mills in a finite volume with the cutoff removed (Balaban, Commun. Math. Phys. 122 (1989) 355 and the series it belongs to; Magnen–Rivasseau–Sénéor, Commun. Math. Phys. 155 (1993) 325, with an infrared cutoff; both metadata). Open: the infinite-volume limit, the axioms in infinite volume, and any spectral information. Jaffe and Witten (p. 7, passage) write that a construction on a compact space alone would be a major step and that no present ideas give a gap uniform in the volume.

7. The list

Statement	Content	Status
T1	finite lattice: unique physical ground state, gap	proved (Theorem 1)
T2	$\Delta_{a,L} = (\hbar c/a)\delta(g; N, G) > 0$ $\delta(g; N, G) \geq c(N)g^2$ for $g \geq g_0(N)$, uniformly in N	Euclidean form established (Osterwalder–Seiler); Hamiltonian form is the next target
T2'	$\delta_\infty(g) > 0$ for all $g > 0$ (no Coulomb phase)	open; false for $U(1)$
T3	$\delta_\infty(g)/(a\Lambda_{\text{lat}}(g)) \rightarrow m/(\hbar c\Lambda_{\text{lat}}) \in (0, \infty)$	open; the Millennium content given T4
T4	continuum infinite-volume theory with the axioms	finite-volume UV stability known; rest open
S	small-volume corner $z(L) = \delta_1 g(L)^{2/3} [1 + O(g^{2/3})]$	C133 plus Lüscher’s expansion

Three of the six rows are theorems or established results; the conjecture is T2' together with T3, given T4. The list replaces “prove the mass gap” by “prove T2 in Hamiltonian form, then push g_0 down”.

8. Consequence for STATE

The next note is T2 in Hamiltonian form: a volume-uniform gap at strong coupling for the Kogut–Susskind Hamiltonian, by a cluster expansion or a relatively bounded perturbation ar-

gument that handles the unbounded electric term. Its output is a number $g_0(N)$; every later step is an improvement of that number toward zero along the scaling curve. The $d = 3$ target of STATE keeps its place as the case where T3 has no transmutation to perform.